MODULE 06 · DOMAIN 1 - SOLUTION DESIGN AND ARCHITECTURE

Business Value Pillars

Efficiency, productivity, transformation, solution cost and performance SLAs: telling them apart and weighing them.

DOMAIN 1 of 7 EXAM WEIGHT 17% MODULE 6 of 38

admin@sthithapragna.com

Time saved is not value until something uses it.

admin@sthithapragna.com

Five pillars: working definitions

PILLAR	WORKING DEFINITION	TYPICAL MEASURE
Efficiency	Less time or effort per task	Minutes per case
Productivity	More valuable output realised	Cases closed per person
Transformation	The operating model changes	New roles, new services
Solution cost	What it costs to run	Cost per transaction
Performance SLA	The service level promised	p95 latency, uptime

ON THE EXAM

Most wrong options name a real benefit under the wrong pillar.

admin@sthithapragna.com

Efficiency or productivity?

EFFICIENCY

Each task is faster

- Measured per task
- Can exist with no business change
- Example: draft in 40 s, not 6 min

PRODUCTIVITY

More is delivered

- Measured in output or outcomes
- Needs freed time put to use
- Example: 30% more cases closed

THE TIEBREAKER

Faster tasks are efficiency. More delivered is productivity. The gap is what people do with the time.

Productivity or transformation?

PRODUCTIVITY

Same process, more output

- The workflow is unchanged
- People do more of the same
- Example: more claims per handler

TRANSFORMATION

A different way of operating

- Roles or processes are redesigned
- Work moves to new kinds of value
- Example: keyers become broker support

THE TIEBREAKER

If the org chart or the service catalogue changed, it is transformation.

Solution cost: count the whole bill

- **Tokens in and out**, at the projected volume, for every call in the flow.
- **Retries and redundant calls**, which quietly multiply spend.
- **Human review and correction time** that the design creates.
- **Infrastructure:** retrieval stores, queues, logging, evaluation runs.

> ON THE EXAM
>
> A cheaper model that doubles human escalations can cost more overall.

Performance SLA: a gate, not a preference

- **A committed level:** metric, threshold, measurement window, consequence.
- **Typical metrics:** p95 latency, availability, an accuracy floor.
- **Fail the SLA and the option is out**, however good its quality or price.
- **Among options that pass,** compare value: cost, efficiency, productivity.

> ON THE EXAM
>
> Check the SLA first. Value only matters among options that meet it.

Attributing observations to pillars

OBSERVATION	PILLAR
Extraction takes 4 minutes instead of 25	Efficiency gain
Underwriters correct fields for 5 extra minutes a pack	Efficiency loss
Each underwriter completes 30% more applications	Productivity
The keying team now runs broker support	Transformation
Spend 40% over budget, from duplicate retries	Solution cost
Agent-assist replies miss the response-time target	Performance SLA

ON THE EXAM

Gains and losses can sit in the same pillar. Net them before you report.

admin@sthithapragna.com

Comparing two architectures

01
Constraints
Regulation, data boundary.

02
SLA gate
Strike any option that fails it.

03
Quality floor
Strike any below the agreed minimum.

04
Cost
Annual, at projected volume, all-in.

05
Value
Efficiency and productivity, net.

06
Recommend
State the trade-offs accepted.

ON THE EXAM

Never trade an SLA breach for quality, and never let a cheap option slip under the quality floor.

Measuring value honestly

- **Baseline before launch:** today's time, output, errors and cost.
- **Measure the whole workflow,** not just the model step.
- **Count review and correction time** against the gain.
- **Leading measures** (time per task) arrive before **lagging ones** (output, revenue).
- **A control group** separates the system's effect from everything else.

> ON THE EXAM
>
> No baseline, no claim. You cannot measure a change you never measured before.

> Three months after launch, Tallyhawk's support report says agents now draft replies in about 40 seconds instead of 6 minutes.
>
> The same report shows the number of tickets each agent resolves per day has not changed.

Which assessment is most accurate?

A Productivity has risen sharply

B The support operation has been transformed

C Solution cost has fallen by the same proportion

D An efficiency gain that has not yet become a productivity gain

◆ PAUSE AND DECIDE

Which assessment is most accurate?

A Productivity has risen sharply

Output per agent is flat. Productivity has not moved.

B The support operation has been transformed

Nothing about roles or the service changed.

C Solution cost has fallen by the same proportion

Drafting time is not running cost; nothing here measures spend.

D An efficiency gain that has not yet become a productivity gain

Faster per task, same output: time saved but not yet used.

THE SIGNAL

Faster tasks with flat output: efficiency without productivity.

> Quillon's portal contract requires replies to begin within 2 seconds at the 95th percentile, and an agreed minimum of 92% correct answers on the evaluation set. Three designs were tested.
>
> Design P: 96% correct, starts at 2.9 s. Design Q: 93% correct, starts at 1.6 s. Design R: 88% correct, starts at 0.9 s and costs least.

Which design should Corvane recommend?

A. Design Q, because it meets both the latency commitment and the accuracy floor

B. Design P, because it is the most accurate, and latency can be tuned after launch

C. Design R, because it is the fastest and the cheapest

D. A mix of P and R by request type, adopted without further testing

◆ PAUSE AND DECIDE

Which design should Corvane recommend?

A Design Q, because it meets both the latency commitment and the accuracy floor

The only design that passes both gates.

B Design P, because it is the most accurate, and latency can be tuned after launch

Fails the contracted latency. A breach is not a tuning task.

C Design R, because it is the fastest and the cheapest

Below the agreed accuracy floor, however cheap.

D A mix of P and R by request type, adopted without further testing

Each half fails a gate, and the mix is untested.

THE SIGNAL

Apply the gates first: the SLA, then the quality floor. Only then compare the survivors.

> A year after Mereside's document extraction went live, its benefits review lists four outcomes.
>
> The review board wants to know which one is a transformation outcome.

Which outcome is transformation?

A. Extracting a document pack is much faster than before

B. API spend came in below budget for the year

C. The keying role was retired and its staff now run a new broker-support service

D. Agent-assist replies met their response-time target all year

◆ PAUSE AND DECIDE

Which outcome is transformation?

A Extracting a document pack is much faster than before

Efficiency: the same task, done faster.

B API spend came in below budget for the year

Solution cost.

C The keying role was retired and its staff now run a new broker-support service

A role removed and a new service created: the operating model changed.

D Agent-assist replies met their response-time target all year

Performance SLA.

THE SIGNAL

A role retired and a new service offered: transformation.

> To save money, Tallyhawk moved its support agent from a capable tier to the lightest tier. Token spend fell by about 60%.
>
> Over the same period, conversations handed to human agents doubled. Each human-handled conversation costs Tallyhawk about USD 9.

What is the most accurate assessment?

A Solution cost must include the extra human handling; the change may have raised total cost

B Solution cost fell by more than half, so the change succeeded

C The extra hand-offs are a performance SLA breach, not a cost question

D The change is a transformation of the support model

◆ PAUSE AND DECIDE

What is the most accurate assessment?

A Solution cost must include the extra human handling; the change may have raised total cost

The whole bill includes the human work the design creates.

B Solution cost fell by more than half, so the change succeeded

Counts tokens only and ignores the doubled hand-offs.

C The extra hand-offs are a performance SLA breach, not a cost question

No SLA is mentioned. The hand-offs carry a clear cost.

D The change is a transformation of the support model

Nothing about roles or services changed.

THE SIGNAL

Cheaper tokens, more human work: solution cost means the whole bill.

Mereside's chief operating officer wants to report the value of agent assist to the board next quarter. Rollout to the contact centre starts in three weeks.

Nobody has recorded current handling times, first-contact resolution or the rate of wrong information given on calls.

What should Corvane do first?

A Survey advisers after launch about how much time they feel they save

B Report token usage and the number of suggestions generated

C Track the percentage of suggestions advisers accept

D Measure today's handling time, first-contact resolution and error rate, and keep a control group during rollout

◆ PAUSE AND DECIDE

What should Corvane do first?

A Survey advisers after launch about how much time they feel they save

Opinion, collected after the fact, with nothing to compare against.

B Report token usage and the number of suggestions generated

Activity, not value.

C Track the percentage of suggestions advisers accept

Acceptance is not correctness or business value.

D Measure today's handling time, first-contact resolution and error rate, and keep a control group during rollout

A baseline and a comparison make a value claim possible.

THE SIGNAL

No baseline, no claim. Measure before the change, and keep a comparison.

> Corvane is preparing the annual business case for Quillon's discharge summaries. The finance director has asked for the solution-cost section only.
>
> Five figures are on the table.

Which two figures belong in the solution-cost section? Choose two.

A Clinician minutes saved per discharge

B Model token spend at the projected annual volume

C Clinician satisfaction scores from the pilot

D Reply latency at the ninety-fifth percentile

E Retrieval, logging and evaluation infrastructure costs

◆ PAUSE AND DECIDE

Which two figures belong in the solution-cost section? Choose two.

A Clinician minutes saved per discharge

Efficiency, a benefit rather than a cost.

B Model token spend at the projected annual volume

The core running cost, at real volume.

C Clinician satisfaction scores from the pilot

Useful adoption evidence, but not a cost.

D Reply latency at the ninety-fifth percentile

Performance SLA.

E Retrieval, logging and evaluation infrastructure costs

Part of the whole bill around the model.

THE SIGNAL

Solution cost is everything it takes to run the system, not only the tokens.

In five lines

- **Efficiency** is faster tasks; **productivity** is more delivered. The gap is what the time is used for.
- **Transformation** changes roles, processes or services.
- **Solution cost** is the whole bill: tokens, retries, review time, infrastructure.
- **The SLA is a gate.** Then the quality floor. Then compare cost and value.
- **No baseline, no claim.** Measure before launch, and keep a comparison.

COMING UP

Domain 1 complete. Next, Domain 2: choosing the model.